Automatic Voice-Controlled Liquid Dispensing System

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1. Introduction

Cooking is one of the parts of the daily routine for most people's lives, and the same goes for those who are visually impaired. However, they often face problems when cooking, such as measuring and pouring liquids, as it can often lead to spills and incorrect amounts. Through the interviews I conducted with visually impaired members of the Vista Center community, one theme that came up multiple times was the ability to measure and transfer liquids reliably, sometimes not just in cooking. One member described pouring soup from one bowl to the other but ended up spilling most of it. These problems create safety hazards like wet floors and make it harder for visually impaired people to cook and live independently.

Existing solutions include the liquid level alarm that notifies a user when a cup is nearly full, but it does not tell the user how much is actually being poured. Automatic dispensers can prevent spills as the user does not need to pour by hand, but most are only designed for water or other beverages rather than for quick transfer/pouring of liquids like soups, oils, and sauces. This means there is still no option that lets users choose how much they want to pour while also preventing overflow.

To address this gap, I am designing a liquid dispensing device that can handle a variety of liquids and can be operated through voice controls. The user can state the amount of liquid they want to dispense, and the device will listen, dispense the quantity the user has indicated, and stop when the target is reached. Additionally, when the liquid is near the rim of the container, the device will stop the dispensing, even if the target amount has not been reached. This will give users control over how much they want to dispense, reduce food waste and safety hazards, and promote independence, especially in kitchens, for people with visual impairments.

2. Background

There are around 2.2 billion people worldwide that have near or distance vision impairment (WHO). In an RNIB survey of 477 blind and partially sighted people in the UK, 40% reported living alone, which meant they needed to complete household tasks independently. Additionally, 60% said being able to happily cook for themselves and others would have a positive impact on their quality of life but only 36% reported that they felt they're definitely able to do so (RNIB, 2023).

Existing Solution 1: RNIB Liquid Level Indicator
This device uses two metal probes that can be hung over the rim of a cup to alert the user when

the cup is nearly full through beeping and vibrating. It has two liquid levels and can help prevent liquid overflowing from containers and spilling. However, the user must still pour and stop manually and the amount poured each time is arbitrary. RNIB Liquid Level Indicator

Existing Solution 2: Cobolt Talking Measuring Jug
The jug audibly says the liquid measurements when a button is pressed or automatically when liquid is being poured. It helps the user measure liquid without being able to see the markings on the jug, which makes it useful for people with visual impairments who cannot read a standard measuring cup. However, the user must still manually add the liquid, which does not help with spill prevention as they must aim into the container. The jug also does not dispense a set amount. Instead, the user must stop pouring at the right marking themselves, so overpouring is still possible if they react too slowly. **Cobolt Talking Measuring Jug**

Motivation

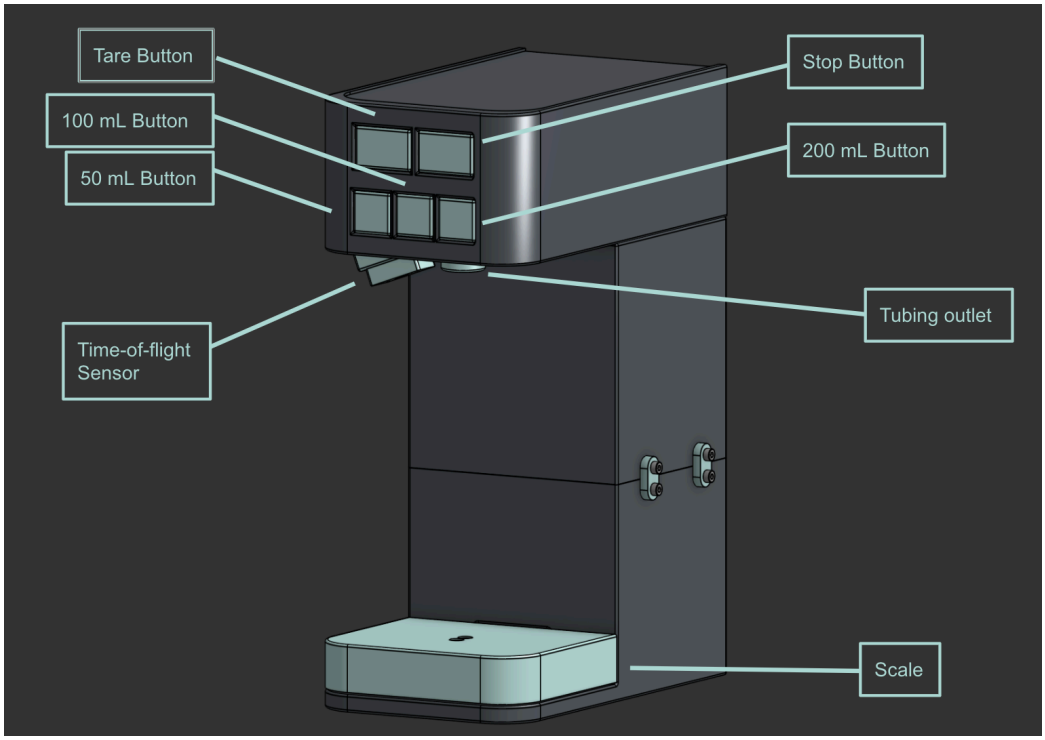
I have seen friends around me, and family members of friends, experience visual impairment and the struggles they face in daily life. I want to make measuring and dispensing liquids easier for people with visual impairments while keeping the solution accessible and affordable. Many existing assistive products are expensive or difficult to find on the market, so I want to create a solution for this problem that can be easily assembled and given to visually impaired people. By making a device that is simple to build and use, I hope to promote their independence and improve their quality of life.

3. System Design

System overview
Explain how the user operates the dispenser, from placing a cup and selecting an amount to completion or stopping. Identify the inputs, controller, sensors, and outputs.

The main parts of the proposed system consists of a central processing unit, a pump to dispense the liquid, a weighing system to measure the amount of liquid dispensed, a time of flight sensor to detect water level, microphone, and buttons with default amounts to dispense and tare and stop.

Figure 1. CAD diagram



Operating Sequence

1. User places cup on scale (doesn't have to be empty)
2. User press tare button to record set starting weight as 0
3. User selects desired amount through either buttons or speaking out loud
4. The correct amount of liquid is transferred into the cup
5. The system measures added weight until the desired target is reached

If at any point during the sequence the time of flight sensor detects that liquid is overflowing, the loop will be stopped and liquid will not be further dispensed.

Hardware Implementation

Control: I used the Raspberry Pi 5 4GB (although a lower-end Raspberry Pi would also work, this was the version I had on hand). It reads the buttons, interprets input from the sensors and microphone, controls the pump, and determines when dispensing should start and stop.

Pump: The pump is powered by a NEMA 17 stepper motor connected to a peristaltic pump, which squeezes liquid through food-grade silicone tubing. A motor driver helps control the pumping mechanism. I initially used an off-the-shelf peristaltic pump (insert link here later), but all the available pumps only supported tubing with a small inner diameter (ID). A small ID limits how much liquid can move through the tube with each rotation, which makes the dispensing process very slow (this is discussed further in the evaluation section). Therefore, I 3D printed my

own peristaltic pump. It has a spacer that allows me to adjust for different sized tubing, and it uses bearing stacks to roll along the tube and push the liquid through.

Weight scale: The weight scale uses a load cell and the HX711 interface to measure the amount of water that is added. The load cell is secured with 3D printed parts and is calibrated using a 100g weight.

Vision sensing: The VL53L5CX sensor measures distance using an 8x8 array of sensing zones. These measurements help determine when the pump should stop dispensing. Because of the sensor's low resolution and difficulty detecting the thin rims of cups, it is mounted above the cup and tilted 30 degrees from horizontal so that it gets a better view of the height of the cup.

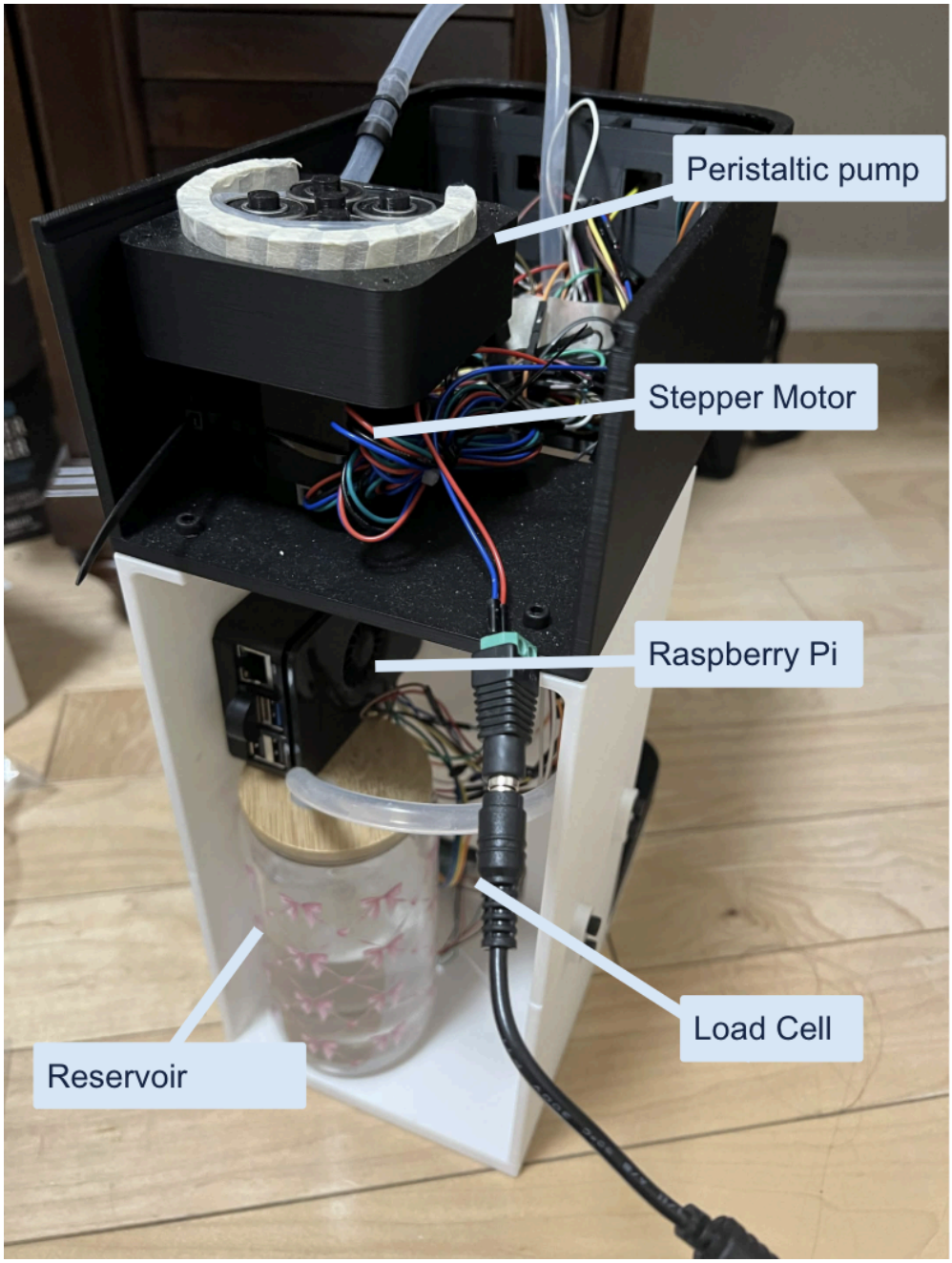
Buttons: There are five buttons on the dispenser: three for the quantities 50, 100, and 200 grams, plus a tare button and a stop button. The tare button zeroes the scale, while the stop button stops the dispensing process. When a quantity button is pressed, the corresponding amount is dispensed.

Microphone: A single KISEER USB microphone is attached to the Raspberry Pi USB port.

Integration: First, I connected each separate component to the Raspberry Pi by itself to test them independently. Then, I combined the pump and the scale so that I could input the amount I wanted to dispense through my program terminal. One end of the food-grade silicone tubing is inserted into a bottle (reservoir) holding the liquid, and the other end is positioned over the cup. Then I added the buttons so that I could use them to control the amount dispensed. Finally, I 3D printed the outer case using PLA, secured each component onto the dispenser system, and added the microphone directly to the Raspberry Pi.

The platform for the weight scale sits beneath the vision sensor and the output end of the silicone tubing. The cup sits on the weight scale. The tubing and components are secured using zip ties and tape.

Figure 2. Hardware implementation



Software implementation

The software uses a finite-state machine to coordinate the user commands. The different states are as follows:

- Untared: The scale has yet to be zeroed
- Taring: The scale averages 30 readings to establish the baseline
- Idle: The system is ready and waiting for a command

- Running: The pump is running while the system is checking the weight to see if target is reached
- Fault states: Pump is stopped because an error occurred

The running (dispensing) state is only allowed after the scale has been tared, which prevents the device from dispensing against an unknown baseline. Separate threads handle motor pulses and voice recognition, allowing these functions to operate alongside continuous scale monitoring.

Weight scale: The HX711 readings are converted into grams through a calibration factor. When taring, the program averages 30 samples to establish the baseline and reduce the influence of random fluctuations in the data. When it is in the dispensing state, the controller calculates the added mass relative to the reading from before the command was registered. The code also includes an experimental linear overshoot-compensation model. This is needed because there is latency between the scale reading and the motor stopping, and water continues to drip out after the motor stops. The model adjusts the cutoff to account for the extra amount delivered.

Vision sensing: The time-of-flight sensor's 8x8 distance grid is processed with a filter. It first rejects readings without a detected target or has invalid data. For each zone, a temporal median filter combines repeated measurements to reduce fluctuation in the input. During validation, each capture contains 30 frames, and a zone is retained only if at least 60% of its readings are valid.

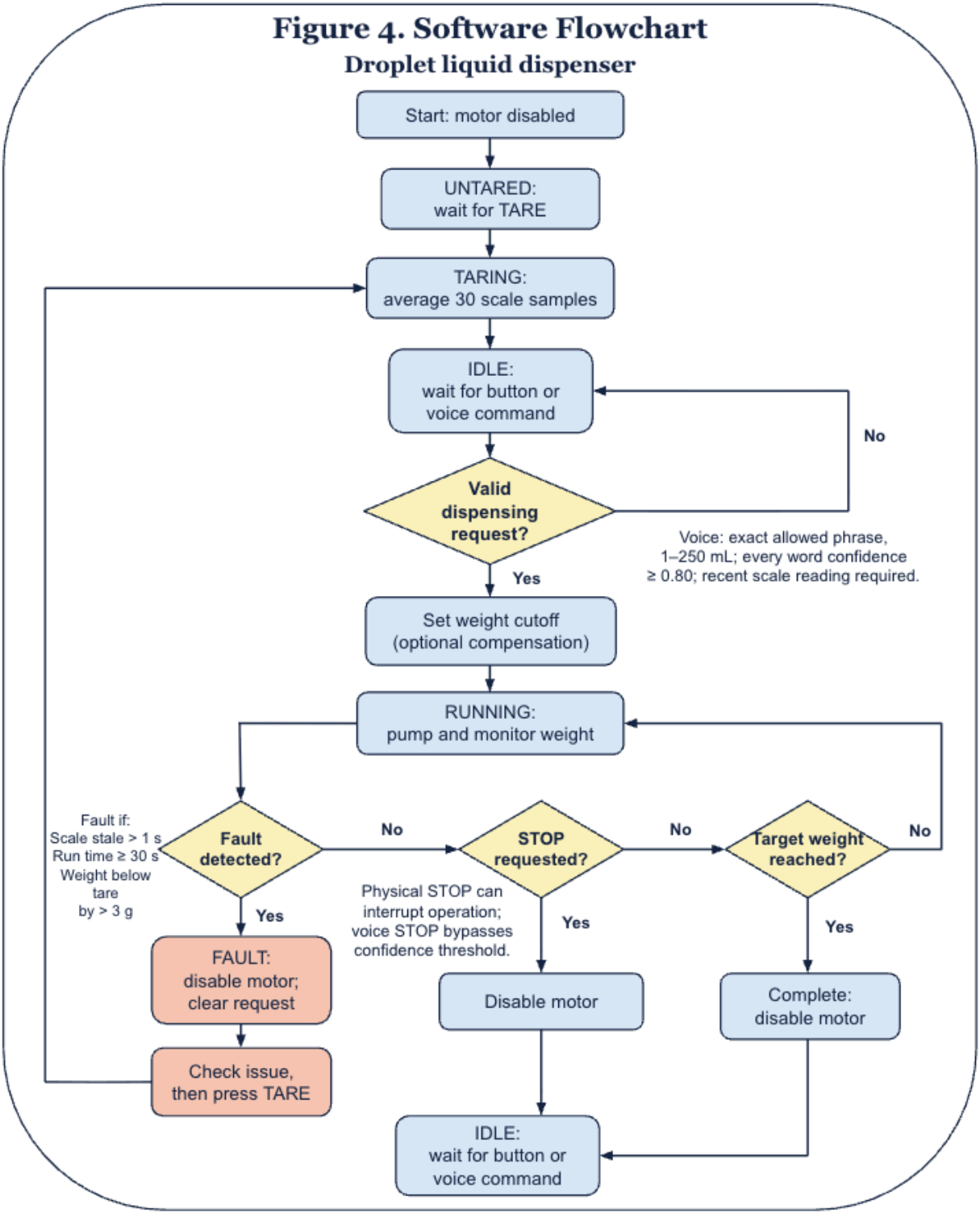
The final grid is classified using weighted nearest-example pattern matching against recorded examples of no cup, empty cup, lower water, middle water, and stop level. These were recorded across multiple trials and set as the baseline for those states. The algorithm also evaluates small horizontal and vertical shifts to accommodate differences in positioning. Weak or ambiguous matches are given the "unknown" label.

Microphone: Offline speech recognition uses Vosk with a restricted vocabulary that only accepts specific commands and numbers within the dispensing range (0 to 250). This system reports a confidence score for each word it recognizes, and the command is only accepted if its confidence is at or above the threshold of 80%. Restricting recognized vocabulary helps to avoid confusion with background noises or conversations.

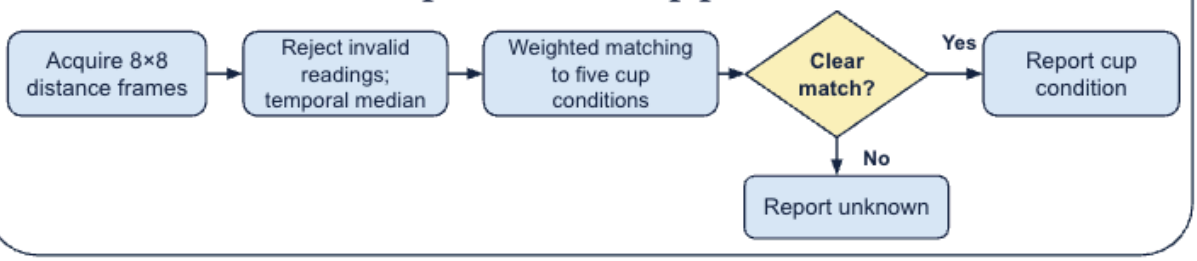
Figure 3. Software flowchart

Figure 4. Software Flowchart

Droplet liquid dispenser



Experimental ToF pipeline



4. Evaluation

Dispensing accuracy: I tested the dispenser using three quantities: 10, 30, and 100 mL. I said the quantity into the microphone and the dispenser poured it. Then I measured the cup on a separate scale and subtracted the starting mass to find the amount added. I used the approximation of 1 g ≈ 1 mL.

I did 15 trials in the first round, five for each amount. The output overshot in every single trial. The average amounts delivered were 13.54 g, 34.36 g, and 111.42 g for the 10, 30, and 100 mL commands.

Because of the overshoots, I added a linear compensation model to adjust the point where the pump stops, and then did another 15 trials with it turned on.

Results

Requested amount	Round 1 trials	Round 1 average absolute error	Round 2 trials	Round 2 average absolute error
10 mL	5	3.54 g	5	0.72 g
30 mL	5	4.36 g	5	0.84 g
100 mL	5	11.42 g	5	1.78 g

The errors were lower at all three quantities in round 2, so the compensation seems to have helped. I'd still want more trials to see if it stays consistent.

Tubing inner diameter: I initially tested with an off-the-shelf peristaltic pump that can only fit 2 mm ID tubing. It took about 12 seconds to dispense 10 mL, so filling 250 mL would take around 5 minutes, which was way too long for everyday use. I then tried 3 mm tubing, but that would still take over 3 minutes for 250 mL. So I moved to a 3D printed peristaltic pump so I could adjust the pump for the tubing size, and I settled on 6 mm ID. After compensation, the average absolute errors were 0.72 g, 0.84 g, and 1.78 g, and I estimate it can dispense 250 mL in about 48 seconds, which is sufficient for my project.

Tubing inner diameter	Time to dispense approximately 10 mL	Flow rate	Estimated time for 250 mL
2 mm	12 seconds	50 mL/min	5 minutes
3 mm	7.7 seconds	78 mL/min	3 minutes 13 seconds

6 mm	1.9 seconds	312 mL/min	48 seconds

The speed of the pump is set in the code by `PULSE_DELAY`, which is currently 0.00001 seconds. It's the delay between the step pulses going to the motor, so a smaller delay makes the motor step faster. But it doesn't keep getting faster forever. At some point the flow rate plateaus. If the motor speed is further increased, it starts stalling which jams the pump. The bigger tubing requires lower motor speeds because it takes more force to squeeze.

Limitations and Improvements: One limitation was that I used the same cup for every trial. Cups with different heights and shapes might change things like how much water splashes out. In some trials I also added water to a cup that already had water in it instead of emptying it, so the starting load wasn't consistent (subtracting the starting mass still gave the amount added, though). Some water also kept dripping from the tubing after the pump stopped, which affects the final amount. Everything I tested was with water, so other liquids would need to be tested too. Further testing should include different cup sizes, more consistent tubing placement, less splashing, and measuring the real flow rate of the 6 mm tubing.

5. Conclusion

This project proposes a voice-controlled automatic liquid dispenser to help visually impaired individuals. The design uses a Raspberry Pi as the main controller, a bearing-based 3D printed peristaltic pump, buttons, a load cell scale, and a time-of-flight sensor to read the liquid level relative to the cup.

Currently, I've built a prototype that can dispense the right amount of water to within about 0.72 g, 0.84 g, and 1.78 g on average for 10, 30, and 100 mL. However, its water level reading ability is not reliable yet. In one of my saved validation runs, it got all three no-cup trials right but returned "unknown" for the empty cup, so I can't give an overall accuracy percentage. The estimated dispense time is around 48 seconds for 250 mL.

Future work can improve the dispensing accuracy, but most importantly the water level reading. A better sensor may be needed because my sensor's resolution was too low and often gave "unknown" instead of actual distance.

6. Resources

CAD model (Onshape):
https://cad.onshape.com/documents/89c1e2d751bea466b80b8828/w/7fbbadfa79b7735ea0154f81/e/75fe2954d9301812b16516c3?renderMode=0&uiState=6ae0259aafaacb436a2ce3e8

Bill of Materials:
https://docs.google.com/spreadsheets/d/1MEt7Rrw82__GPrZ_N5E3bGAwAnIcnz3e56m892PVnfU/edit?usp=sharing

Code: https://github.com/coolFalcon-pixel/dispenser

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